Trustworthy Machine Learning

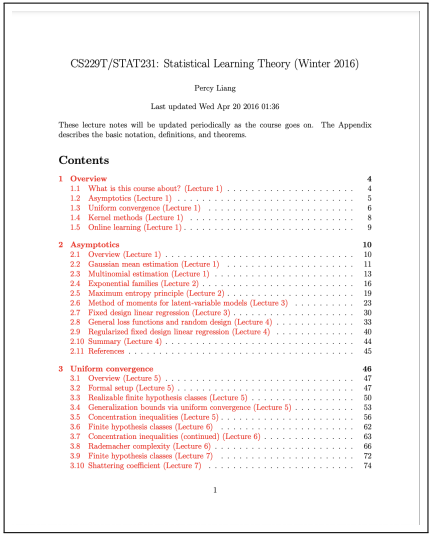
Beyond PAC Learning

Sangdon Park

POSTECH

September 11, 2023

Contents from



CS229T/STAT231: Statistical Learning Theory (Winter 2016)

Percy Liang

Last updated Wed Apr 20 2016 01:36

These lecture notes will be updated periodically as the course goes on. The Appendix describes the basic notation, definitions, and theorems.

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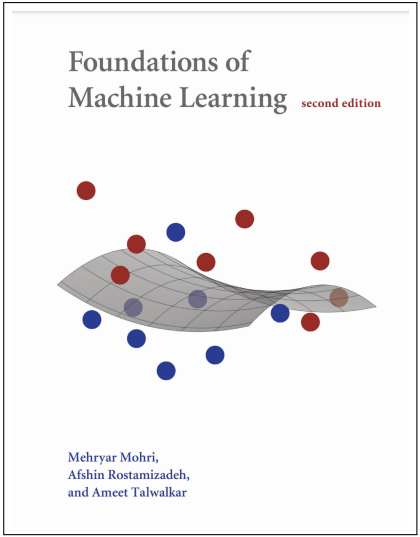


Foundations of Machine Learning **second edition**

Mehryar Mohri,
Afshin Rostamizadeh,
and Ameet Talwalkar

and various papers.

Is PAC Learning Okay?

Four Ingredients of Learning:

- Distribution $\mathcal{D}$
- Loss ℓ
- Hypothesis Space $\mathcal{H}$
- A Learning Algorithm $\mathcal{A}$

Problem?

Is PAC Learning Okay?

Four Ingredients of Learning:

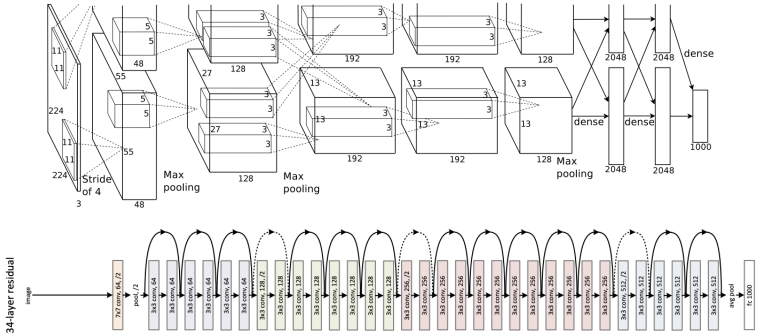
- Distribution $\mathcal{D}$
- Loss ℓ
- Hypothesis Space $\mathcal{H}$
- A Learning Algorithm $\mathcal{A}$

Problem?

The main assumption of PAC learning: $\mathcal{D}$ is separable by some $h^* \in \mathcal{H}$.

$\mathcal{D}$ Is Generally Not Separable

Usually we do not know a set of hypotheses $\mathcal{H}$ that has the true hypothesis h^*.



- What is the architecture of neural networks that perfectly classifies ImageNet?
- We mainly search for good hypothesis space $\mathcal{F}$ without any assumption on $\mathcal{D}$.

Contents

1 Concentration Inequalities

2 Generalization Bounds via Uniform Convergence

Contents

1 Concentration Inequalities

2 Generalization Bounds via Uniform Convergence

Why Concentration Inequalities?

- Understanding the expected loss is a key in statistical learning

$$\min_{f \in \mathcal{F}} \mathbb{E}\ell(x, y, f)$$

- Concentration inequalities
  - Powerful tools that show an appropriate combination of independent random variables will *concentrate* around its *expectation*.
  - A concentration inequality provides a bound around an expected value.
- An Example: Mean estimation
  - Let $X_1, \ldots, X_n$ be i.i.d. real-valued random variables with mean $\mu := \mathbb{E}[X_1]$
  - The empirical mean is defined as

$$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^{n} X_i$$

  - What is the relation between μ and $\hat{\mu}_n$?

Possible Arguments I

Consistency: Due to the law of large numbers,

$$\hat{\mu}_n - \mu \xrightarrow{P} 0$$

- $\xrightarrow{P}$: convergence "in probability"
- If we get more data, $\hat{\mu}_n$ reaches to μ

✗ Asymptotic guarantee: it does not answer on the required number of samples to reach to the correct answer.

Possible Arguments II

Asymptotic normality: Assuming $\text{Var}(X_1) = \sigma^2$, due to the central limit theorem,

$$\sqrt{n}(\hat{\mu}_n - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma^2)$$

- $\xrightarrow{D}$: convergence "in distribution"
- If we get more data, $\hat{\mu}_n$ reaches to μ, where the variance is decreasing at a rate of $1/n$.

✗ Asymptotic guarantee: it does not answer on the required number of samples to reach to the correct answer.

Possible Arguments III

Tail bound: we wish to have a statement as follows:

$$\mathbb{P}\left\{|\hat{\mu}_n - \mu| \geq \varepsilon\right\} \leq \text{SomeFunctionOf}(n, \varepsilon) = \delta.$$

- ε: a desired error level
- $1 - \delta$: the confidence of the error statement

✓ "SomeFunctionOf$(n, \varepsilon) = \delta$" provides the required number of samples to reach a desired level of error with a desired level of confidence.

Hoeffding's Inequality

Theorem

Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in [a_i, b_i]$ for all $i \in \{1, \ldots, n\}$. Then, for any $\varepsilon > 0$, the following inequalities hold for $S_n := \sum_{i=1}^n X_i$:

$$\mathbb{P}\left\{\mathbb{E}\{S_n\} - S_n \geq \varepsilon\right\} \leq \exp\left\{\frac{-2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right\}$$

- Why is it called a tail bound?
- What's the effect of n?
- $X_1, \ldots, X_n$ need not to follow the same distribution

Binomial Distribution Tail Bound

A special version of the Hoeffding's inequality.

Theorem

Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in \{0, 1\}$ and $\mathbb{P}\{X_i = 1\} = p \in [0, 1]$ for all $i \in \{1, \ldots, n\}$. Then, for any $\varepsilon > 0$, the following inequalities hold for $S_n = \sum_{i=1}^{n} X_i$:

$$\mathbb{P}\left\{p \leq \hat{p}\right\} \geq 1 - \delta,$$

where $F(k; n, p)$ is the CDF of a binomial distribution with n trials and success probability p and $\hat{p} := \inf\left\{p' \in [0, 1] \mid F(S_n; n, p') \leq \delta\right\}$.

- This is from the Clopper-Pearson interval for estimating binomial confidence intervals.
- $\hat{p}$ is the "best" value that fails to "describe" our observation S_n.
- From the Hoeffding's inequality, $\mathbb{P}\left\{\frac{S_n}{n} - p > \varepsilon\right\} \leq \exp\left\{-2n\varepsilon^2\right\}$

McDiarmid's Inequality

A generalized version of the Heoffding's inequality.

Theorem

Let $(X_1, \ldots, X_n) \in \mathcal{X}^n$ be a list of $n \geq 1$ independent random variables and assume that there exist $c_1, \ldots, c_n > 0$ such that $f : \mathcal{X}^n \to \mathbb{R}$ satisfies the following conditions:

$$\left| f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_m) \right| \leq c_i,$$

for all $i \in \{1, \ldots, n\}$ and any $x_1, \ldots, x_n, x_i \in \mathcal{X}$. Let $f(S)$ denote $f(X_1, \ldots, X_n)$, then, for all $\varepsilon > 0$, the following inequality holds:

$$\mathbb{P}\left\{ f(S) - \mathbb{E}\{f(S)\} \geq \varepsilon \right\} \leq \exp\left\{ \frac{-2\varepsilon^2}{\sum_{i=1}^n c_i^2} \right\}.$$

- Useful concentration inequality for a more complex function than a mean value under the "bounded difference".
- The main inequality for a generalization bound.

Contents

1 Concentration Inequalities

2 Generalization Bounds via Uniform Convergence

Agnostic PAC Learning Algorithm

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Toward Efficient Agnostic Learning

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Editor: Lisa Hellerstein

Abstract. In this paper we initiate an investigation of generalizations of the Probably Approximately Correct (PAC) learning model that attempt to significantly weaken the target function assumptions. The ultimate goal in this direction is informally termed *agnostic learning*, in which we make virtually no assumptions on the target function. The name derives from the fact that as designers of learning algorithms, we give up the belief that Nature (as represented by the target function) has a simple or succinct explanation. We give a number of positive and negative results that provide an initial outline of the possibilities for agnostic learning. Our results include hardness results for the most obvious generalization of the PAC model to an agnostic setting, an efficient and general agnostic learning method based on dynamic programming, relationships between loss functions for agnostic learning, and an algorithm for a learning problem that involves hidden variables.

Keywords: machine learning, agnostic learning, PAC learning, computational learning theory

- For the smooth transition from PAC learning, I will introduce agnostic PAC learning.
- Later, we will mainly use languages from statistical learning theory.

Agnostic PAC Learning Algorithm

Definition (simplified definition)

An algorithm $\mathcal{A}$ is an agnostic PAC-learning algorithm for $\mathcal{H}$ if for any $\varepsilon > 0$, $\delta > 0$, ~~$h^* \in \mathcal{H}$~~, and $\mathcal{D}$ ~~separable by h^*~~, and for some minimum sample size n^* (which depends on $\varepsilon, \delta, \mathcal{D}$), the following holds with any sample size $n \geq n'$:

$$\mathbb{P}\left\{L(\mathcal{A}(\mathcal{S})) - \min_{h \in \mathcal{H}} L(h) \leq \varepsilon\right\} \geq 1 - \delta,$$

where $\mathcal{S} := ((x_1, y_1), \ldots, (x_n, y_n)) \sim \mathcal{D}^n$.

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- $\min_{h \in \mathcal{H}} L(h)$: the best hypothesis

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- Vapnik notations on generalization bounds are more widely used.

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- $\min_{h \in \mathcal{H}} L(h)$: the best hypothesis
- Vapnik notations on generalization bounds are more widely used.
- Please check out the original agnostic PAC learning definition.

Example: A Learning Bound for a Finite Hypothesis Set I

Setup:

- $\mathcal{H}$: a finite set of functions mapping from $\mathcal{X}$ to $\mathcal{Y}$
- $\mathcal{D}$: any distribution — no assumption!
- $\mathcal{S}$: labeled examples
- $\mathcal{A}$: any algorithm — no assumption to use!

Example: A Learning Bound for a Finite Hypothesis Set II

Theorem

For any $\varepsilon > 0$, $\delta > 0$, and $\mathcal{D}$, we have

$$\forall h \in \mathcal{H}, \quad L(h) \leq \hat{L}(h) + \sqrt{\frac{\log|\mathcal{H}| + \log\frac{2}{\delta}}{2n}}$$

with probability at least $1 - \delta$.

- We have logarithmic dependence on $|\mathcal{H}|$ and $1/\delta$ – this bound is not "sensitive" to them.
- This is a uniform convergence bound: "$\forall h$" is inside of the probability.

$$(\text{✗}) \quad \forall h \in \mathcal{H}, \quad \mathbb{P}\left\{L(h) \leq \hat{L}(h) + \sqrt{\frac{\log|\mathcal{H}| + \log\frac{2}{\delta}}{2n}}\right\} \geq 1 - \delta$$

- Conservative (=data-independent): even though some h is "bad", we need the convergence guarantee.

Example: A Learning Bound for a Finite Hypothesis Set III

Proof Sketch:

$$
\begin{aligned}
\mathbb{P}\left\{\exists h \in \mathcal{H},\ L(h) - \hat{L}(h) > \varepsilon\right\} &= \mathbb{P}\left\{\bigvee_{h \in \mathcal{H}} L(h) - \hat{L}(h) > \varepsilon\right\} \\
&\leq \sum_{h \in \mathcal{H}} \mathbb{P}\left\{L(h) - \hat{L}(h) > \varepsilon\right\} && (1) \\
&\leq |\mathcal{H}| \exp\left\{-2n\varepsilon^2\right\} && (2)
\end{aligned}
$$

- (1): Uniform convergence via the union bound
- (2): a "point" convergence via the Hoeffding's inequality

Separable $\mathcal{D}$ v.s. $\mathcal{D}$

A bound under the separability assumption

$$L(\mathcal{A}(\mathcal{S})) \leq \frac{1}{n}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right)$$

A bound without separability

$$\forall h \in \mathcal{H}, \quad L(h) \leq \hat{L}(h) + \sqrt{\frac{\log|\mathcal{H}| + \log\frac{1}{\delta}}{2n}}$$

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- A bound that exploits more information is tighter.
  - A distribution is separable ($\approx$ no noise).
  - The expected error is the parameter of a Bernoulli distribution.

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- A bound that exploits more information is tighter.
  - A distribution is separable ($\approx$ no noise).
  - The expected error is the parameter of a Bernoulli distribution.
- Under the additional information, we can learn faster (*i.e.*, $\frac{1}{n}$ vs $\frac{1}{\sqrt{n}}$).

A More General Bound

- In general, $\mathcal{H}$ is infinite (*e.g.*, a set of neural networks)

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- Related keywords include
  - McDiarmid's Inequality
  - Rademacher Complexity
  - VC dimension
  - A learning bound for SVM

A More General Bound

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- The related bound is one of the key results of statistical learning theory (via Vapnik)
- Related keywords include
  - McDiarmid's Inequality
  - Rademacher Complexity
  - VC dimension
  - A learning bound for SVM
- Caution: this "data-independent" bound cannot not explain the learnability of deep networks!

Definitions

Definition (best hypothesis)

$$h^* := \min_{h \in \mathcal{H}} L(h)$$

Definition (empirical risk minimizer)

$$\hat{h} := \min_{h \in \mathcal{H}} \hat{L}(h)$$

Goal: Find Generalization Bounds

An Interesting Quantity:

$$L(h) - \hat{L}(h)$$

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$$L(\hat{h}) - L(h^*)$$

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  - It is bounded as follows:

$$\mathbb{P}\left\{L(\hat{h}) - L(h^*) \geq \varepsilon\right\} \leq \mathbb{P}\left\{\sup_{h \in \mathcal{H}} \left|L(h) - \hat{L}(h)\right| \geq \frac{\varepsilon}{2}\right\}$$

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  - I'll introduce the philosophy on "From Theory to Algorithm", where $L(h) - \hat{L}(h)$ is more directly related.

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  - We also call a bound of $L(h) - \hat{L}(h)$ a generalization bound.
  - I'll introduce the philosophy on "From Theory to Algorithm", where $L(h) - \hat{L}(h)$ is more directly related.
- The generalization bound will depend on the complexity of $\mathcal{H}$, which is harder to measure if $\mathcal{H}$ is an infinite set (than the finite case).

Rademacher Complexity

A way to measure the complexity of $\mathcal{H}$!

Definition

Let $\mathcal{F}$ be a set of real-valued functions $f : \mathcal{Z} \to \mathbb{R}$ (*e.g.*, $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$). The Rademacher complexity of $\mathcal{F}$ is

$$R_n(\mathcal{F}) := \mathbb{E}\left\{\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i f(Z_i)\right\},$$

where $Z_1, \ldots, Z_n$ are drawn i.i.d. from a distribution and $\sigma_1, \ldots, \sigma_n$ are drawn i.i.d. from the uniform distribution over $\{-1, +1\}$ (a.k.a. Rademacher variables).

- Previously, "concentration inequalities" + "union bound provides" a generalization bound.
- This term will be upper-bounded by a term with "VC dimension" later.

Rademacher Complexity: Interpretation

$$R_n(\mathcal{F}) := \mathbb{E}\left\{\sup_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i f(Z_i)\right\}$$

- This term considers an "imaginary binary classification" problem with randomly labeled examples (Z_i, σ_i).
  - If $\sigma_i = \mathrm{sign}(f(Z_i))$, f is correct on (Z_i, σ_i).
  - Solving $\sup$ = finding a "best" binary classifier.
  - Fix n and $\mathcal{F} \to$ draw Z_i and $\sigma_i \to$ find f.
- $R_n(\mathcal{F})$ captures how well the "best classifier" from $\mathcal{F}$ can align with random labels.
  - Large $R_n(\mathcal{F})$ means that there is some $f \in \mathcal{F}$, "flexible" enough to learn randomly labeled examples.
  - *e.g.*, linear functions v.s. neural networks
- We wish to find $\mathcal{F}$ such that $R_n(\mathcal{F}) \to 0$ as $n \to \infty$.

Generalization Bound via Rademacher Complexity

Theorem

Let $\mathcal{F} := \{z \mapsto \ell(z, h) \mid h \in \mathcal{H}\}$ *and* $\ell(\cdot) \in [0, 1]$. *For all* $h \in \mathcal{H}$,

$$L(h) \leq \hat{L}(h) + 2R_n(\mathcal{F}) + \sqrt{\frac{\ln \frac{1}{\delta}}{2n}}$$

with probability at least $1 - \delta$.

- $f \in \mathcal{F}$ is a composition of h and ℓ.

Proof Sketch: A Bird's-eye View

1. Define a random variable G_n
   - $G_n := \sup_{h \in \mathcal{H}} L(h) - \hat{L}(h)$
   - A maximum difference between the expected and empirical error.
   - The bound of this term is a generalization bound.

Proof Sketch: A Bird's-eye View

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   - We will use McDiarmid's inequality.

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   - A maximum difference between the expected and empirical error.
   - The bound of this term is a generalization bound.
2. Show that G_n concentrates to $\mathbb{E}\{G_n\}$.
   - We will use McDiarmid's inequality.
3. Use a technique called "symmetrization" to bound $\mathbb{E}\{G_n\}$ using the Rademacher complexity.

Proof Sketch

1. Setup

Define an interesting quantity to us!

- Consider the maximum difference between $L(h)$ and $\hat{L}(h)$.

$$G_n := \sup_{h \in \mathcal{H}} L(h) - \hat{L}(h)$$

  - G_n is a random variable that depends on $Z_1, \ldots, Z_n$.
  - $L(\hat{h}) - \hat{L}(\hat{h})$ is what we want to understand and $G_n \geq L(\hat{h}) - \hat{L}(\hat{h})$ due to $\sup$.

- We will consider the following tail bound:

$$\mathbb{P}\left\{G_n \geq \varepsilon\right\}.$$

Proof Sketch I

2. Concentration

Convert a tail bound into an expectation!

- Let g be the deterministic function such that $G_n \coloneqq g(Z_1, \ldots, Z_n)$.
- Then, the following holds:

$$\left| g(Z_1, \ldots, Z_i, \ldots, Z_n) - g(Z_1, \ldots, Z_i', \ldots, Z_n) \right| \leq \frac{1}{n}.$$

- Why?
  - Recall $\hat{L}(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(Z_i, h)$.
  - Recall $\ell(\cdot) \in [0, 1]$.
  - We have

$$\left| \underbrace{\sup_{h \in \mathcal{H}} \left[L(h) - \hat{L}(h) \right]}_{g(Z_1, \ldots, Z_i, \ldots, Z_n)} - \underbrace{\sup_{h \in \mathcal{H}} \left[L(h) - \hat{L}(h) + \frac{1}{n} \left(\ell(Z_i, h) - \ell(Z_i', h) \right) \right]}_{g(Z_1, \ldots, Z_i', \ldots, Z_n)} \right| \leq \frac{1}{n}.$$

Proof Sketch II

2. Concentration

- Apply the McDiarmid's inequality:

$$\mathbb{P}\left\{G_n \geq \mathbb{E}\{G_n\} + \varepsilon'\right\} \leq \exp\left(-2n\varepsilon'^2\right).$$

  - g is a non-trivial function, including $\sup$ over $h \in \mathcal{H}$; thus, we cannot use the usual concentration inequality (*e.g.*, the Hoeffding's inequality).
  - But, we can still use the McDiarmid's inequality due to the bounded difference.

Proof Sketch I

3. Symmetrization

Bound $\mathbb{E}\{G_n\}$!

- $\mathbb{E}\{G_n\}$ is not easy to analysis as it depends on $L(h)$, an expectation of an unknown distribution $\mathcal{D}$.
- We will replace this to depend on $\mathcal{D}$ only through samples $Z_1, \ldots, Z_n$.
- The key idea of "symmetrization" is to introduce "ghost" samples $Z'_1, \ldots, Z'_n$, drawn i.i.d. from $\mathcal{D}$ to rewrite $\mathbb{E}\{G_n\}$.
  - Let $\hat{L}'(h) := \frac{1}{n}\sum_{i=1}^{n} \ell(Z'_i, h)$.
  - Rewrite $L(h)$ in terms of the ghost samples, *i.e.,*

$$\mathbb{E}\{G_n\} = \mathbb{E}\left\{\sup_{h\in\mathcal{H}} L(h) - \hat{L}(h)\right\} = \mathbb{E}\left\{\sup_{h\in\mathcal{H}} \mathbb{E}\{\hat{L}'(h)\} - \hat{L}(h)\right\}$$

Proof Sketch II

3. Symmetrization

- Simplify and bound this rewritten $\mathbb{E}\{G_n\}$:

$$
\begin{aligned}
\mathbb{E}\{G_n\} &= \mathbb{E}\left\{\sup_{h\in\mathcal{H}} \mathbb{E}\{\hat{L}'(h)\} - \hat{L}(h)\right\} \\
&= \mathbb{E}\left\{\sup_{h\in\mathcal{H}} \mathbb{E}\left\{\hat{L}'(h) - \hat{L}(h) \;\middle|\; Z_{1:n}\right\}\right\} \\
&\leq \mathbb{E}\left\{\mathbb{E}\left\{\sup_{h\in\mathcal{H}} \hat{L}'(h) - \hat{L}(h) \;\middle|\; Z_{1:n}\right\}\right\} \\
&= \mathbb{E}\left\{\sup_{h\in\mathcal{H}} \hat{L}'(h) - \hat{L}(h)\right\} \\
&= \mathbb{E}\left\{\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell(Z_i', h) - \ell(Z_i, h)\right)\right\}
\end{aligned}
$$

Proof Sketch III

3. Symmetrization

- Remove the dependence on the ghost samples.
  - Introduce the i.i.d. Rademacher variables $\sigma_1, \ldots, \sigma_n$, where σ_i is uniform over $\{-1, 1\}$.
  - Observe that $\ell(Z_i', h) - \ell(Z_i, h)$ is symmetric around 0.
  - Thus, we have

$$
\begin{aligned}
\mathbb{E}\{G_n\} &\leq \mathbb{E}\left\{\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell(Z_i', h) - \ell(Z_i, h)\right)\right\} \\
&= \mathbb{E}\left\{\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i \left(\ell(Z_i', h) - \ell(Z_i, h)\right)\right\} \\
&\leq \mathbb{E}\left\{\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i \ell(Z_i', h) + \sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} (-\sigma_i) \ell(Z_i, h)\right\} \\
&= 2\mathbb{E}\left\{\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i \ell(Z_i, h)\right\} = 2R_n(\mathcal{F})
\end{aligned}
$$

Proof Sketch

4. Combine

- From concentration, we have

$$\mathbb{P}\left\{G_n \geq \mathbb{E}\{G_n\} + \varepsilon'\right\} \leq \exp\left(-2n\varepsilon'^2\right).$$

- From symmetrization, we have

$$\mathbb{E}\{G_n\} \leq 2R_n(\mathcal{F}).$$

- Our goal is to bound the following tail probability:

$$\begin{aligned}\mathbb{P}\{G_n \geq \varepsilon\} &\leq \exp\left(-2n\left(\varepsilon - \mathbb{E}\{G_n\}\right)^2\right) \\ &\leq \exp\left(-2n\left(\varepsilon - 2R_n(\mathcal{F})\right)^2\right)\end{aligned}$$

- This shows the claim, as

$$\delta = \exp\left(-2n\left(\varepsilon - 2R_n(\mathcal{F})\right)^2\right) \quad \Rightarrow \quad \varepsilon = 2R_n(\mathcal{F}) + \sqrt{\frac{\ln\frac{1}{\delta}}{2n}}.$$